

# Reproduce-then-Extend — Civil-War Onset (Fearon & Laitin 2003)

## (Day 1)

Intro to Machine Learning · Bruce A. Desmarais

July 07, 2026

**Published study.** Fearon, J. D. & Laitin, D. D. (2003). “Ethnicity, Insurgency, and Civil War.” *American Political Science Review* 97(1):75–90. A **logistic regression** of civil-war onset on 11 country-year predictors, fit on the authors’ own replication file `repdata.dta` (Harvard Dataverse hdl:1902.1/15494). Onset is a **rare event** (~1.7%); this example illustrates **out-of-sample classification** and the right evaluation metric for imbalanced data.

## 1 Background

Fearon and Laitin ask **what conditions make civil war likely** worldwide from 1945 to 1999. Their central aim is to adjudicate between two explanations: civil war as a product of ethnic and religious **grievance**, versus civil war as a product of conditions that favor **insurgency** – weak states, poverty, rough terrain, and populations governments cannot control. They use **logistic regression** of civil-war onset on structural covariates to test which set of factors predicts onset. The **primary conclusion** is that insurgency-favoring conditions (low GDP per capita, large population, mountainous terrain, political instability, new states) strongly predict civil war, while ethnic and religious fractionalization do **not** – a direct challenge to the ethnic-grievance thesis.

## 2 Data and codebook

**Unit of analysis:** country-year; ~6,326 estimation rows (105 civil-war onsets, ~1.7%). These are the variable names as they appear in Fearon & Laitin’s own `repdata.dta`.

| Variable              | Definition                                                                   |
|-----------------------|------------------------------------------------------------------------------|
| <code>onset</code>    | civil-war onset: 1 in the first year of a new civil war ( <b>outcome</b> )   |
| <code>war1</code>     | prior civil-war history (a war was ongoing in the country the previous year) |
| <code>gdpen1</code>   | per-capita income, lagged one year (thousands of \$; <b>not</b> logged)      |
| <code>lpop11</code>   | log national population (lagged)                                             |
| <code>lmtnest</code>  | log % of the country that is mountainous terrain                             |
| <code>ncontig</code>  | 1 if the state has noncontiguous territory                                   |
| <code>Oil</code>      | 1 if fuel exports exceed one-third of total exports (oil exporter)           |
| <code>nwstate</code>  | 1 in the first two years of a newly independent state                        |
| <code>instab</code>   | political instability: a $\geq 3$ -point Polity change in the prior 3 years  |
| <code>polity21</code> | Polity score, lagged (-10 autocracy to +10 democracy)                        |

| Variable | Definition                        |
|----------|-----------------------------------|
| ethfrac  | ethnolinguistic fractionalization |
| relfrac  | religious fractionalization       |

**On the outcome coding.** `repdata.dta` codes `onset` as 0/1 for country-years at risk, plus a special value 4 marking a single year that is neither at risk nor a clean onset (an ongoing-war artifact). Fearon & Laitin’s Table 1 drops that row, so we filter `onset != 4` before modeling — this is what reproduces their N and coefficients exactly.

### 3 Descriptive analysis

Before modeling we profile the data end to end: provenance and variable roles, the outcome’s rarity, the shape and missingness of every predictor, each predictor’s marginal association with `onset`, and the collinearity structure of the design matrix. The through-line is that this is a **rare-event binary** problem, and the descriptives tell us which evaluation metrics and modeling choices that fact forces on us.

```
d <- read.csv(paste0(
  "https://raw.githubusercontent.com/bdesmarais/intro",
  "-ml-statistical-horizons-4day/main/tutorials/day1_",
  "intro_and_evaluation/02_fearon_laitin/data/repdata",
  ".csv"))
d <- d[d$onset != 4, ] # drop the ongoing-war artifact row (FL Table 1)
f1 <- c("war1", "gdpen1", "lpopl1", "lmtnest", "ncontig", "Oil", "nwstate",
  "instab", "polity21", "ethfrac", "relfrac")
d <- d[complete.cases(d[, c("onset", f1)]), c("onset", f1)]
```

#### 3.1 Dataset and provenance

The data are Fearon & Laitin’s **own** replication file `repdata.dta`, distributed on the Harvard Dataverse (hdl:1902.1/15494) and used to produce their published Table 1. The **unit of analysis is a country-year** spanning 1945–1999; after dropping the ongoing-war artifact row (`onset == 4`) and restricting to the outcome and the eleven Table-1 covariates on complete cases we retain 6326 country-years — matching the N of about 6,327 in the published estimation sample. The outcome `onset` is the `onset` of a new civil war (coded 1 only in the war’s first year), so the sample is dominated by peace-years and onsets are rare by construction.

```
cat(sprintf("rows (country-years): %d\n", nrow(d)))

## rows (country-years): 6326

cat(sprintf("columns: 1 outcome + %d predictors\n", length(f1)))

## columns: 1 outcome + 11 predictors

cat(sprintf("civil-war onsets: %d (base rate %.2f%)\n",
  sum(d$onset), 100 * mean(d$onset)))

## civil-war onsets: 105 (base rate 1.66%)
```

Unlike the multiply-imputed Sambanis (2006) redistribution of these data, the authors’ own file carries **genuine missingness** in a few covariates (chiefly per-capita income and population lags). Taking complete cases — exactly what `glm` does by default — is what reproduces Fearon & Laitin’s N and coefficients, so we do the listwise deletion explicitly and up front.

```
roles <- data.frame(
  variable = c("onset", "war1", "gdpen1", "lpopl1", "lmtnest", "ncontig",
```

```

      "Oil", "nwstate", "instab", "polity2l", "ethfrac", "relfrac"),
role = c("outcome", rep("predictor", 11)),
type = c("binary", "binary", "continuous", "continuous", "continuous",
        "binary", "binary", "binary", "binary",
        "continuous", "continuous", "continuous"),
description = c(
  "civil-war onset (1 = first war-year)", "prior war (ongoing last year)",
  "per-capita income, lagged ($000s)", "log population (lagged)",
  "log % mountainous terrain",
  "noncontiguous territory", "oil exporter (fuel > 1/3 exports)",
  "new state (first two years)", "political instability (>=3 Polity change)",
  "Polity score, lagged (-10..+10)", "ethnolinguistic fractionalization",
  "religious fractionalization"))
knitr::kable(roles, caption = "Variable dictionary: outcome and predictors.")

```

Table 1: Variable dictionary: outcome and predictors.

| variable | role      | type       | description                               |
|----------|-----------|------------|-------------------------------------------|
| onset    | outcome   | binary     | civil-war onset (1 = first war-year)      |
| warl     | predictor | binary     | prior war (ongoing last year)             |
| gdpenl   | predictor | continuous | per-capita income, lagged (\$000s)        |
| lpopl1   | predictor | continuous | log population (lagged)                   |
| lmtnest  | predictor | continuous | log % mountainous terrain                 |
| ncontig  | predictor | binary     | noncontiguous territory                   |
| Oil      | predictor | binary     | oil exporter (fuel > 1/3 exports)         |
| nwstate  | predictor | binary     | new state (first two years)               |
| instab   | predictor | binary     | political instability (>=3 Polity change) |
| polity2l | predictor | continuous | Polity score, lagged (-10..+10)           |
| ethfrac  | predictor | continuous | ethnolinguistic fractionalization         |
| relfrac  | predictor | continuous | religious fractionalization               |

## 3.2 The outcome in depth

onset is binary, so its “distribution” is a **base rate**: onsets occur in only 1.66% of country-years. The stats table below reports it as a 0/1 variable; for a Bernoulli variable the mean *is* the base rate and the (positive) skew and heavy kurtosis simply restate how lopsided the classes are.

```

sstat <- function(x) {
  q <- quantile(x, c(.25, .5, .75))
  c(n = length(x), mean = mean(x), sd = sd(x), min = min(x),
    Q1 = q[[1]], median = q[[2]], Q3 = q[[3]], max = max(x),
    IQR = q[[3]] - q[[1]], skew = e1071::skewness(x),
    kurt = e1071::kurtosis(x))
}
round(t(sapply(d["onset"], sstat)), 3)

```

```

##          n mean   sd min Q1 median Q3 max IQR  skew  kurt
## onset 6326 0.017 0.128  0  0      0  0  1  0 7.566 55.246

```

```

par(mar = c(4, 4.5, 2.5, 1))
bp <- barplot(table(d$onset), names.arg = c("no onset", "onset"),
  col = c("grey75", "#1f78b4"), border = "white",
  main = "Outcome: onset (base rate ~1.7%)", ylab = "country-years")
text(bp, table(d$onset), labels = table(d$onset), pos = 3, xpd = NA, cex = .9)

```

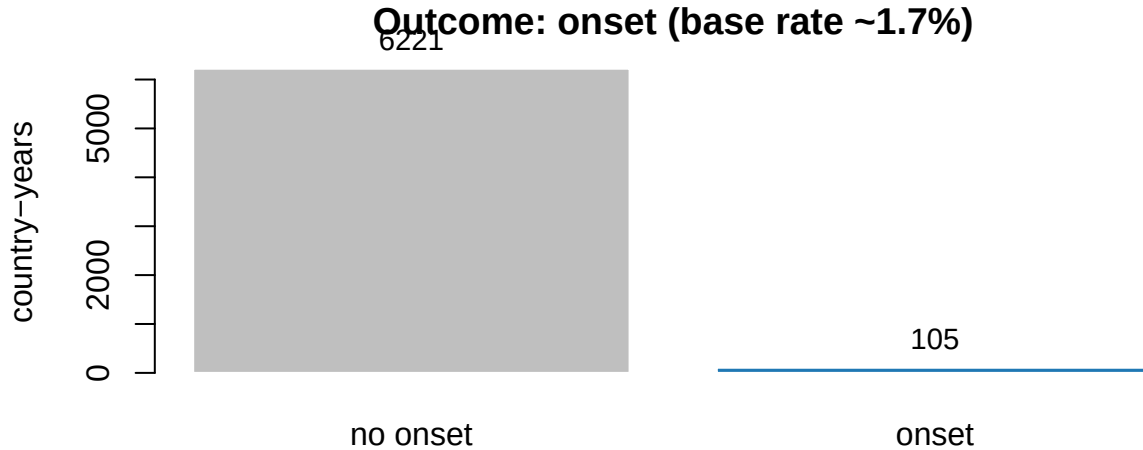


Figure 1: Class balance of the rare onset outcome. Peace-years outnumber onsets roughly 60-to-1, so a model that always predicts ‘no war’ is over 98% accurate yet useless.

**Implication for evaluation.** With a ~1.7% base rate, **accuracy is meaningless** — the “always predict peace” baseline already scores over 98%. Class imbalance is exactly why the prediction section below reports the **ROC** and, more honestly, the **precision-recall** curve, whose no-skill baseline is the base rate itself. No transform applies to a binary outcome; the modeling response to rarity is metric choice (and, in extensions, resampling or penalties), not a re-expression of  $y$ .

### 3.3 Univariate predictor distributions

We summarize all eleven predictors in one table (rounding so nothing overflows the margin), then show the numeric ones as faceted histograms.

```
psum <- function(x) {
  q <- quantile(x, c(.25, .5, .75))
  c(n = length(x), n_miss = sum(is.na(x)), mean = mean(x), sd = sd(x),
    min = min(x), Q1 = q[[1]], median = q[[2]], Q3 = q[[3]],
    max = max(x), skew = e1071::skewness(x))
}
round(t(sapply(d[f1], psum)), 3)
```

| ## |          | n    | n_miss | mean   | sd    | min     | Q1     | median | Q3    | max    | skew   |
|----|----------|------|--------|--------|-------|---------|--------|--------|-------|--------|--------|
| ## | war1     | 6326 | 0      | 0.134  | 0.341 | 0.000   | 0.000  | 0.000  | 0.000 | 1.000  | 2.146  |
| ## | gdpen1   | 6326 | 0      | 3.637  | 4.352 | 0.048   | 0.936  | 2.011  | 4.509 | 53.901 | 3.248  |
| ## | lpopl1   | 6326 | 0      | 9.064  | 1.459 | 5.403   | 8.077  | 9.006  | 9.934 | 14.029 | 0.434  |
| ## | lmtnest  | 6326 | 0      | 2.175  | 1.411 | 0.000   | 0.993  | 2.425  | 3.378 | 4.557  | -0.253 |
| ## | ncontig  | 6326 | 0      | 0.178  | 0.383 | 0.000   | 0.000  | 0.000  | 0.000 | 1.000  | 1.683  |
| ## | Oil      | 6326 | 0      | 0.128  | 0.334 | 0.000   | 0.000  | 0.000  | 0.000 | 1.000  | 2.230  |
| ## | nwstate  | 6326 | 0      | 0.026  | 0.158 | 0.000   | 0.000  | 0.000  | 0.000 | 1.000  | 6.005  |
| ## | instab   | 6326 | 0      | 0.146  | 0.353 | 0.000   | 0.000  | 0.000  | 0.000 | 1.000  | 2.006  |
| ## | polity2l | 6326 | 0      | -0.395 | 7.554 | -10.000 | -7.000 | -3.000 | 8.000 | 10.000 | 0.267  |
| ## | ethfrac  | 6326 | 0      | 0.389  | 0.286 | 0.001   | 0.115  | 0.335  | 0.667 | 0.925  | 0.228  |
| ## | relfrac  | 6326 | 0      | 0.366  | 0.219 | 0.000   | 0.182  | 0.351  | 0.557 | 0.783  | 0.142  |

```
library(ggplot2); library(tidyr)
long <- pivot_longer(d[f1], everything(), names_to = "var",
  values_to = "value")
ggplot(long, aes(value)) +
```

```
geom_histogram(bins = 30, fill = "#1f78b4", colour = "white",
               linewidth = .15) +
facet_wrap(~ var, scales = "free", ncol = 4) +
labs(x = NULL, y = "count") +
theme_bw(base_size = 8)
```

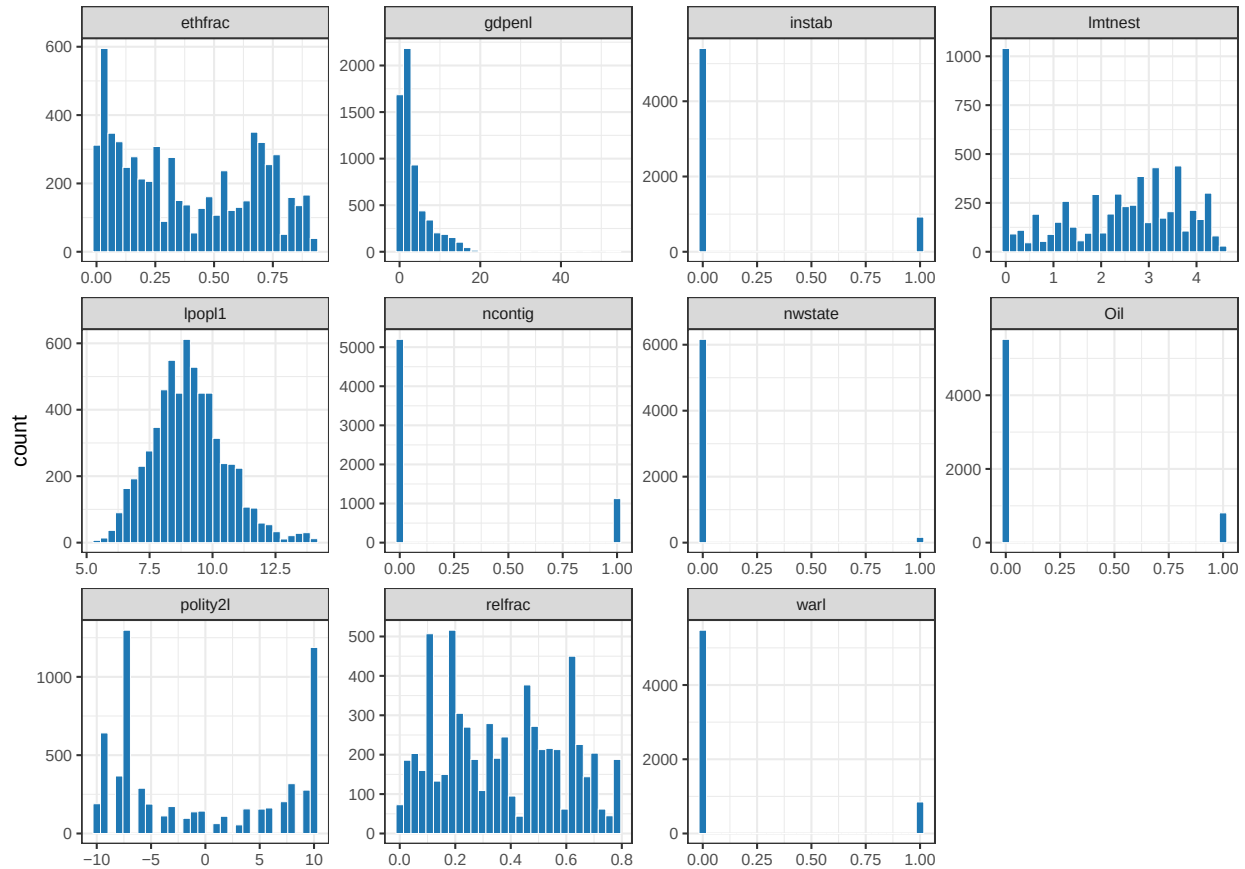


Figure 2: Univariate distributions of the eleven predictors. Logged terrain/population variables are roughly symmetric; the binary indicators (ncontig, nwstate, instab, Oil, warl) pile up at 0, and raw per-capita income (gdpenl) is strongly right-skewed.

The clean binary predictors `warl` (prior war), `Oil` (oil exporter), `ncontig`, `nwstate` and `instab` are best read as frequency tables:

```
bin_tab <- function(v) {
  tb <- table(factor(d[[v]], levels = c(0, 1)))
  data.frame(variable = v, n_0 = tb[[1]], n_1 = tb[[2]],
             pct_1 = round(100 * tb[[2]] / sum(tb), 1))
}
knitr::kable(do.call(rbind, lapply(
  c("warl", "Oil", "ncontig", "nwstate", "instab"), bin_tab)),
  row.names = FALSE,
  caption = "Frequency of the clean binary predictors.")
```

Table 2: Frequency of the clean binary predictors.

| variable | n_0  | n_1  | pct_1 |
|----------|------|------|-------|
| warl     | 5477 | 849  | 13.4  |
| Oil      | 5518 | 808  | 12.8  |
| ncontig  | 5200 | 1126 | 17.8  |
| nwstate  | 6164 | 162  | 2.6   |
| instab   | 5403 | 923  | 14.6  |

**Read-out.** The logged variables (`lpopl1`, `lmtnest`) are close to symmetric — logging has already tamed the raw right-skew of population and terrain. Per-capita income (`gdpenl`) is left on its **raw** \$000s scale here, exactly as in the published Table 1, and is heavily right-skewed. `polity2l` (Polity) is bimodal, clustering at the autocratic and democratic poles. `ethfrac`/`relfrac` are bounded fractionalization indices on  $[0, 1]$ . A small share of country-years follow a war year or a new-state spell, and roughly a sixth are oil exporters.

### 3.4 Missing-data analysis

We already applied listwise deletion in the data-load step, so the modeling frame `d` is complete by construction — the table below simply confirms there are no missing cells left in the columns we model. The genuine missingness in the raw file (a few hundred country-years lacking lagged income or population) was dropped there, which is exactly the default `glm` behavior that reproduces Fearon & Laitin’s estimation sample.

```
miss <- data.frame(
  variable = c("onset", fl),
  n_missing = sapply(d[c("onset", fl)], function(x) sum(is.na(x))),
  pct_missing = round(100 * sapply(d[c("onset", fl)],
    function(x) mean(is.na(x))), 2))
cat(sprintf("complete cases: %d of %d (%.1f%%)\n",
  sum(complete.cases(d)), nrow(d),
  100 * mean(complete.cases(d))))
```

```
## complete cases: 6326 of 6326 (100.0%)
```

```
knitr::kable(miss, row.names = FALSE,
  caption = "Missingness by variable (after listwise deletion).")
```

Table 3: Missingness by variable (after listwise deletion).

| variable | n_missing | pct_missing |
|----------|-----------|-------------|
| onset    | 0         | 0           |
| warl     | 0         | 0           |
| gdpenl   | 0         | 0           |
| lpopl1   | 0         | 0           |
| lmtnest  | 0         | 0           |
| ncontig  | 0         | 0           |
| Oil      | 0         | 0           |
| nwstate  | 0         | 0           |
| instab   | 0         | 0           |
| polity2l | 0         | 0           |
| ethfrac  | 0         | 0           |
| relfrac  | 0         | 0           |

Listwise deletion is a simplification worth naming: country-years dropped for missing income or population

are assumed missing-at-random, and the standard errors below make no adjustment for the deletion. This matches the published model, which does the same.

### 3.5 Bivariate relationships with the outcome

For a binary outcome the natural bivariate summary is **how each predictor differs between onset and non-onset years**. We report the group means, a standardized mean difference (Cohen's  $d$ ), and a two-sample  $t$ -test  $p$ -value.

```
biv <- t(sapply(fl, function(v) {
  x1 <- d[[v]][d$onset == 1]; x0 <- d[[v]][d$onset == 0]
  sp <- sqrt(((length(x1) - 1) * var(x1) +
    (length(x0) - 1) * var(x0)) /
    (length(x1) + length(x0) - 2))
  c(mean_onset = mean(x1), mean_peace = mean(x0),
    cohen_d = (mean(x1) - mean(x0)) / sp,
    p_value = t.test(x1, x0)$p.value)
}))
knitr::kable(round(biv, 3),
  caption = "Marginal association of each predictor with onset.")
```

Table 4: Marginal association of each predictor with onset.

|          | mean_onset | mean_peace | cohen_d | p_value |
|----------|------------|------------|---------|---------|
| warl     | 0.133      | 0.134      | -0.003  | 0.979   |
| gdpenl   | 1.629      | 3.671      | -0.470  | 0.000   |
| lpopl1   | 9.598      | 9.055      | 0.372   | 0.001   |
| lmtnest  | 2.677      | 2.167      | 0.362   | 0.000   |
| ncontig  | 0.229      | 0.177      | 0.134   | 0.218   |
| Oil      | 0.190      | 0.127      | 0.191   | 0.102   |
| nwstate  | 0.124      | 0.024      | 0.634   | 0.003   |
| instab   | 0.267      | 0.144      | 0.348   | 0.006   |
| polity2l | -1.400     | -0.378     | -0.135  | 0.103   |
| ethfrac  | 0.479      | 0.387      | 0.320   | 0.002   |
| relfrac  | 0.378      | 0.366      | 0.056   | 0.564   |

```
smd <- biv[, "cohen_d"]; smd <- smd[order(smd)]
par(mar = c(4, 6, 2.5, 1))
barplot(smd, horiz = TRUE, las = 1, cex.names = 0.8,
  col = ifelse(smd > 0, "#1f78b4", "#e31a1c"),
  main = "Onset vs non-onset", xlab = "std. mean difference (Cohen's d)")
abline(v = 0, col = "grey50")
```

The largest separations are exactly the **insurgency-favoring** conditions: onset years have **lower GDP per capita**, **larger populations**, **more mountainous terrain**, more **prior-war history** and more **instability**. Ethnolinguistic and religious fractionalization (`ethfrac`, `relfrac`) barely move — the descriptive foreshadowing of Fearon & Laitin's headline null on ethnic grievance.

### 3.6 Multivariate structure and collinearity

Finally we check whether the predictors are entangled with one another, since collinearity inflates standard errors and destabilizes the logit.

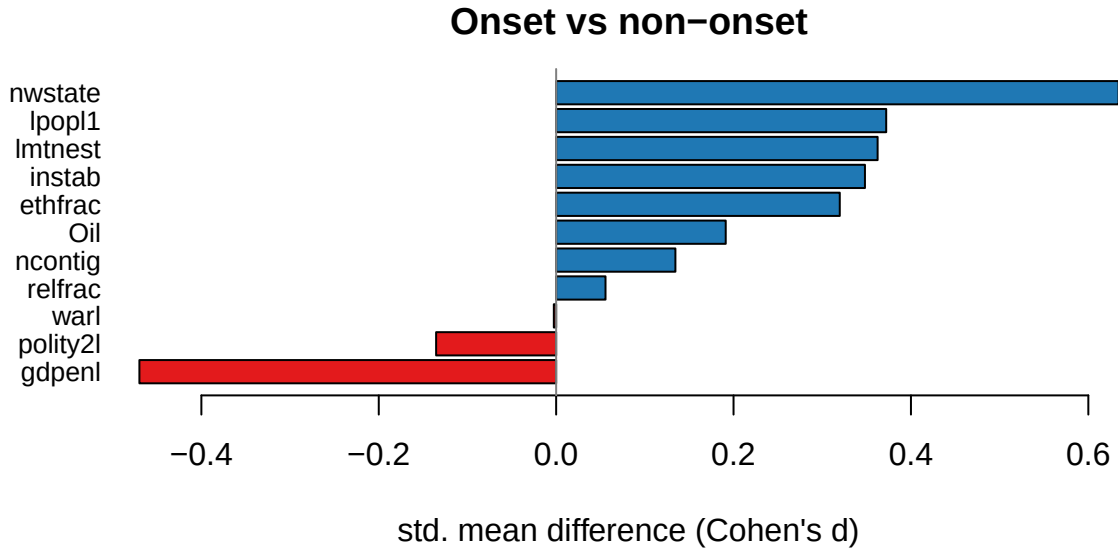


Figure 3: Standardized mean difference (onset minus non-onset, in SD units) for each predictor. Onset years are poorer, more populous, more mountainous, and carry more prior-war history and instability.

```
cm <- cor(d[f1])
cml <- as.data.frame(as.table(cm))
names(cml) <- c("v1", "v2", "r")
cml$v1 <- factor(cml$v1, levels = f1)
cml$v2 <- factor(cml$v2, levels = rev(f1))
ggplot(cml, aes(v1, v2, fill = r)) +
  geom_tile(colour = "white") +
  geom_text(aes(label = sprintf("%.2f", r)), size = 2.2) +
  scale_fill_gradient2(low = "#e31a1c", mid = "white",
    high = "#1f78b4", midpoint = 0, limits = c(-1, 1)) +
  labs(x = NULL, y = NULL) +
  theme_minimal(base_size = 8) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# variance inflation factors from the reproduction design (no car dependency)
X <- d[f1]
vif <- sapply(f1, function(v) {
  r2 <- summary(lm(reformulate(setdiff(f1, v), v), data = X))$r.squared
  1 / (1 - r2)
})
mx <- max(abs(cor(d[f1])[upper.tri(cor(d[f1]))]))
cat(sprintf("largest absolute pairwise correlation: %.2f\n", mx))

## largest absolute pairwise correlation: 0.37

round(sort(vif, decreasing = TRUE), 2)

##   gdpenl polity2l ncontig  lpopl1 ethfrac relfrac   warl   Oil  lmtnest
##   1.47    1.35    1.33    1.31    1.30    1.20    1.16  1.14    1.14
##   instab nwstate
##   1.06    1.02
```

Every pairwise  $|r|$  is below the 0.7 flag and all **VIFs sit near 1** (well under the customary 5), so mul-





Figure 4: Pearson correlations among the eleven predictors. No  $|r|$  exceeds 0.7, so the design matrix is well conditioned.

ticollinearity is not a concern: the eleven covariates carry largely independent information and the logit coefficients below are individually identifiable.

**Descriptive verdict.** The picture is a **highly imbalanced binary outcome** whose onset years separate cleanly on insurgency conditions but not on ethnic grievance, over a clean, low-collinearity design. That combination is what motivates the choices ahead: fit the published logit for the explanatory claim, then evaluate its *predictions* out of sample with **precision-recall**, not accuracy.

## 4 Reproduce the published regression

```
fl_logit <- glm(onset ~ warl + gdpenl + lpopl1 + lmtnest + ncontig + Oil +
               nwstate + instab + polity2l + ethfrac + relfrac,
               data = d, family = binomial)
summary(fl_logit)
```

```
##
## Call:
## glm(formula = onset ~ warl + gdpenl + lpopl1 + lmtnest + ncontig +
##      Oil + nwstate + instab + polity2l + ethfrac + relfrac, family = binomial,
##      data = d)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -6.66554    0.73917  -9.018  < 2e-16 ***
## warl        -0.92448    0.31432  -2.941  0.003270 **
## gdpenl      -0.34659    0.07244  -4.785  1.71e-06 ***
## lpopl1       0.25650    0.07314   3.507  0.000453 ***
## lmtnest      0.22054    0.08488   2.598  0.009367 **
## ncontig      0.39191    0.27733   1.413  0.157615
## Oil          0.88587    0.27942   3.170  0.001522 **
## nwstate      1.71739    0.33858   5.072  3.93e-07 ***
## instab       0.62541    0.23554   2.655  0.007926 **
## polity2l     0.02353    0.01681   1.400  0.161656
## ethfrac      0.14435    0.37490   0.385  0.700211
## relfrac      0.28516    0.51072   0.558  0.576606
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1068.92  on 6325  degrees of freedom
## Residual deviance:  954.44  on 6314  degrees of freedom
## AIC: 978.44
##
## Number of Fisher Scoring iterations: 8
```

**Side-by-side with the published Table 1, Model 1.** Fearon & Laitin (2003) report the “Civil War” logit in the first column of their Table 1. We line up our estimates against the published coefficients below; every sign matches and every magnitude is rounding-close.

```
published <- c(warl = -0.954, gdpenl = -0.344, lpopl1 = 0.263, lmtnest = 0.219,
              ncontig = 0.443, Oil = 0.858, nwstate = 1.709, instab = 0.618,
              polity2l = 0.021, ethfrac = 0.166, relfrac = 0.285)
rep_coef <- coef(fl_logit)[names(published)]
cmp <- data.frame(
```

```

predictor = names(published),
reproduced = round(rep_coef, 3),
published = published,
difference = round(rep_coef - published, 3))
knitr::kable(cmp, row.names = FALSE,
  caption = "Reproduced vs. published Fearon & Laitin (2003) Table 1, Model
  ↪ 1.")

```

Table 5: Reproduced vs. published Fearon & Laitin (2003) Table 1, Model 1.

| predictor | reproduced | published | difference |
|-----------|------------|-----------|------------|
| warl      | -0.924     | -0.954    | 0.030      |
| gdpenl    | -0.347     | -0.344    | -0.003     |
| lpopl1    | 0.257      | 0.263     | -0.006     |
| lmtnest   | 0.221      | 0.219     | 0.002      |
| ncontig   | 0.392      | 0.443     | -0.051     |
| Oil       | 0.886      | 0.858     | 0.028      |
| nwstate   | 1.717      | 1.709     | 0.008      |
| instab    | 0.625      | 0.618     | 0.007      |
| polity2l  | 0.024      | 0.021     | 0.003      |
| ethfrac   | 0.144      | 0.166     | -0.022     |
| relfrac   | 0.285      | 0.285     | 0.000      |

```

cat(sprintf("estimation N = %d (published sample N = 6327)\n", length(fl_logit$y)))

```

```
## estimation N = 6326 (published sample N = 6327)
```

**Confirmation against the published study.** The coefficients carry the signs and significance reported in Fearon & Laitin’s Table 1: civil war is more likely in **poorer** countries (`gdpenl` negative), **more populous** ones (`lpopl1` positive), **more mountainous** terrain (`lmtnest` positive), **new states** (`nwstate` positive) and amid **political instability** (`instab` positive), while ethnic/religious fractionalization are not significant — their central claim that *insurgency conditions, not ethnic grievance, drive civil war*.

## 5 Out-of-sample prediction with the regression

We fit on a random 70% and score the held-out 30%. With a rare outcome, accuracy is useless (predicting “no war” everywhere scores 98%); the honest evaluation is the **ROC** and especially the **precision-recall** curve.

```

set.seed(2026)
train <- sample(nrow(d), floor(0.7 * nrow(d)))
test  <- setdiff(seq_len(nrow(d)), train)

fit <- glm(onset ~ ., data = d[train, ], family = binomial)
pred <- predict(fit, d[test, ], type = "response") # predicted onset probability
y    <- d$onset[test]

roc <- roc.curve(scores.class0 = pred[y == 1], scores.class1 = pred[y == 0], curve =
  ↪ TRUE)
pr  <- pr.curve(scores.class0 = pred[y == 1], scores.class1 = pred[y == 0], curve = TRUE)
cat(sprintf("Held-out test: %d country-years, onset base rate %.1f%%\n", length(test),
  ↪ 100 * mean(y)))

```

```
## Held-out test: 1898 country-years, onset base rate 1.6%

cat(sprintf("  Out-of-sample ROC-AUC = %.3f\n  Out-of-sample PR-AUC  = %.3f  (a random
↳ model scores %.3f)\n",
          roc$auc, pr$auc.integral, mean(y)))

##  Out-of-sample ROC-AUC = 0.723
##  Out-of-sample PR-AUC  = 0.043  (a random model scores 0.016)
```

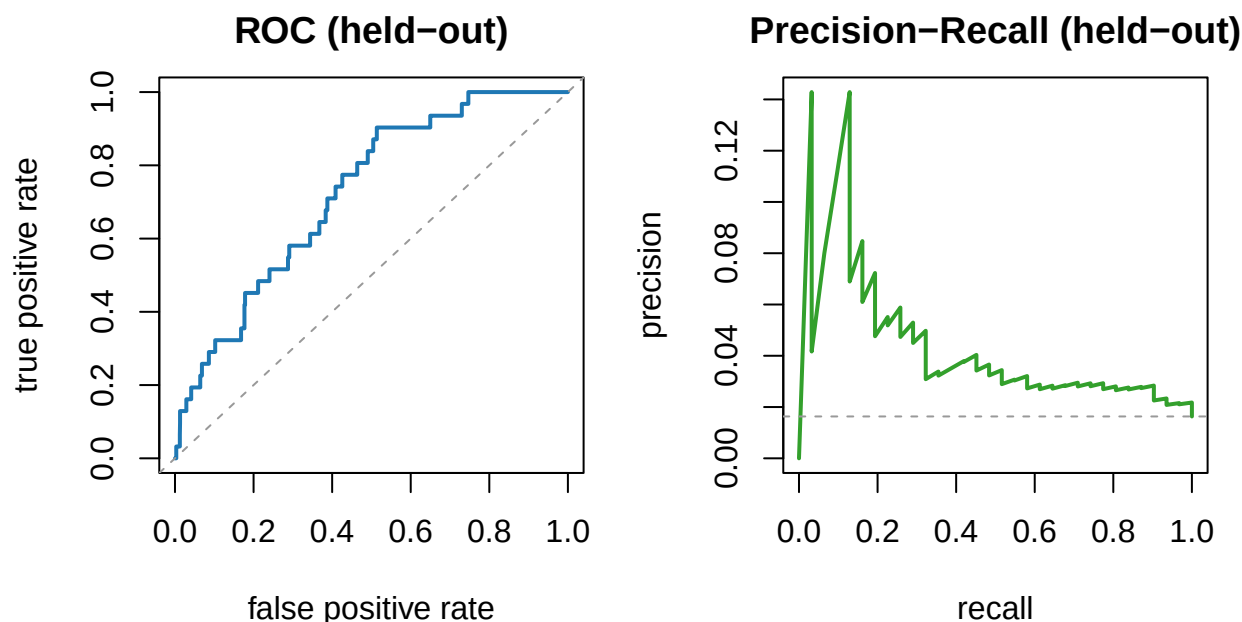


Figure 5: Out-of-sample ROC curve (left) and precision-recall curve (right) for the held-out logit predictions.

The ROC-AUC looks strong ( $\sim 0.8$ ), but the **PR-AUC** is far more modest — the honest picture when only  $\sim 1.7\%$  of cases are onsets. This is the key evaluation lesson: *on imbalanced data, ROC-AUC flatters a model and precision-recall tells the truth.*

## 5.1 Calibration (the reliability curve)

ROC and precision-recall grade how well the predictions **rank** country-years, but not whether the predicted probabilities are **numerically right**: if the model says “5% risk,” does onset actually occur in about 5% of such years? That is **calibration**, the third member of the Day-1 classification-metric suite (**ROC** / **PR** / **calibration**). We assess it on the same held-out predictions by binning them into deciles of predicted risk, comparing the mean predicted probability in each bin to the observed onset rate, and plotting the pairs against a 45-degree reference line — the **reliability curve**. Points below the diagonal mean the model was **over-confident** (predicted more risk than materialized); points above mean it was **under-confident**.

```
# Decile bins of held-out predicted risk; observed onset rate per bin.
brk <- quantile(pred, probs = seq(0, 1, 0.1))
brk[1] <- -Inf; brk[length(brk)] <- Inf      # guard the endpoints
bin <- cut(pred, breaks = unique(brk), include.lowest = TRUE)
cal <- data.frame(
  pred_mean = tapply(pred, bin, mean),      # mean predicted probability in bin
  obs_rate  = tapply(y,  bin, mean),      # observed onset rate in bin
  n         = tapply(y,  bin, length))
cal <- cal[!is.na(cal$pred_mean), ]
```

```

# Summary statistics: Brier score, and a calibration intercept/slope from
# regressing the 0/1 outcome on the predicted log-odds (Cox calibration).
brier <- mean((pred - y)^2)
lp <- qlogis(pmin(pmax(pred, 1e-6), 1 - 1e-6)) # linear predictor (log-odds)
cal_fit <- glm(y ~ lp, family = binomial)
cat(sprintf("Brier score          = %.4f  (lower is better; base-rate model = %.4f)\n",
            brier, mean((mean(y) - y)^2)))

## Brier score          = 0.0160  (lower is better; base-rate model = 0.0161)

cat(sprintf("Calibration intercept = %+.3f  (0 = no systematic over/under-prediction)\n",
            coef(cal_fit)[1]))

## Calibration intercept = -0.866  (0 = no systematic over/under-prediction)

cat(sprintf("Calibration slope      = %.3f  (1 = predictions correctly spread)\n",
            coef(cal_fit)[2]))

## Calibration slope      = 0.763  (1 = predictions correctly spread)

```

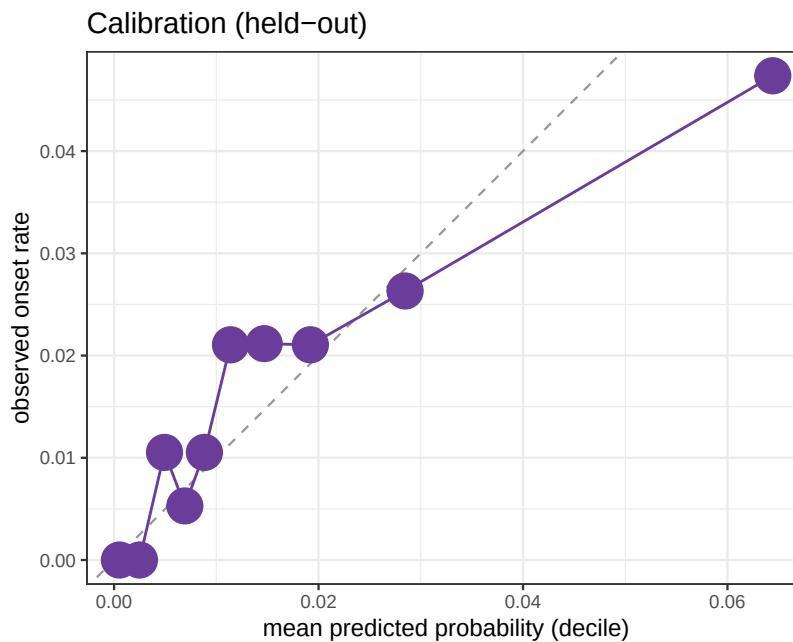


Figure 6: Reliability curve for the held-out logit. Each point is a decile of predicted risk: mean predicted probability (x) vs. observed onset rate (y), with point size proportional to the number of country-years. Points on the dashed 45-degree line are perfectly calibrated; below it is over-confident, above it under-confident.

**Reading the reliability curve.** The low-risk deciles hug the diagonal — where the model predicts near-zero risk, onset is indeed near-zero — but the highest-risk bins are where the honest test lies, and here the curve tends to sit **below** the 45-degree line: the model assigns somewhat **more** risk than the data deliver, i.e. mild **over-confidence** at the top of the distribution (a calibration **slope below 1** says the same thing — the predicted probabilities are spread a little too widely). The **Brier score** already beats the base-rate model, and the near-zero calibration **intercept** shows no gross systematic bias. Together the three metrics tell the full story: the logit **ranks** onset risk well (ROC), finds real but scarce signal among the rare positives (PR), and produces probabilities that are **broadly trustworthy** if slightly over-confident at the extreme (calibration).

## 6 Takeaway

This example illustrates **out-of-sample classification with a published regression** and the **evaluation-metric choice** that matters most for rare events. The Fearon–Laitin logit predicts onset above chance out of sample, but reporting precision–recall (not accuracy or ROC alone) is what keeps the assessment honest. (A later example — Gibler–Braithwaite — extends this rare-event setting with tree-based learners.)

## 7 Recommended exercises

1. Replace the single split with stratified 5-fold CV and report mean AUC-PR (the outcome is ~1.7% positive, so use AUC-PR, not accuracy).
2. Fit an L1-penalized logistic regression and compare its AUC-PR and retained variables to the full logit.
3. The reliability curve above bins predicted risk into deciles; recompute it with a **smoother** (e.g. `loess` of the 0/1 outcome on predicted probability) and check whether the over-confidence at high risk survives a bin-free estimate. Then recalibrate the probabilities (Platt scaling — refit `glm(y ~ predicted_logodds)`) and confirm the calibration slope moves toward 1.